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Modeling and validation of reaction wheel micro-vibrations considering imbalances and bearing disturbances

Hassan Alkomy, Jinjun Shan*

Department of Earth and Space Science and Engineering, York University, 4700 Keele St., Toronto, Canada, M3J 1P3

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A B S T R A C T

Reaction wheels are one of the main actuators for spacecraft attitude control. However, they generate undesired micro-vibrations that affect the attitude control performance. Some space missions that have high-resolution instruments such as telescopes and cameras require high-accuracy attitude control. In such missions, these micro-vibrations cause stability problems, which negatively affect the performance of these instruments. This paper introduces a nonlinear dynamics model for these micro-vibrations in five degrees of freedom. The main advantage of the proposed model is that it analytically considers various disturbance sources. This allows the reaction wheel designers to predict the micro-vibration behavior of the reaction wheel even before the manufacturing of the reaction wheel, *i.e.*, in the design stage. The model considers static and dynamic imbalances of the wheel and the bearing disturbances that come from the waviness in all bearing parts (inner/outer races and balls). Experiments are conducted to validate the effectiveness of the nonlinear model.

1. Introduction

Nowadays, space missions require high-accuracy attitude control, which, in turn, requires better understanding of the micro-vibrations induced by spacecraft subsystems. Reaction wheel (RW) is the main actuator for spacecraft attitude control. At the same time, it is one of the main disturbance sources of onboard micro-vibrations.

Obtaining the dynamics model of a reaction wheel that describes its micro-vibration behavior depends on the method used in the derivation and the required model accuracy. Many models have been developed and can be categorized based on the method used to derive the dynamics model. Generally speaking, there are three methods: the empirical models, the combined models, and the analytical models.

The empirical models depend mostly on experiments. Basically, the disturbance behavior is assumed to be a series of sinusoidal forces added to the basic equations of motion. These sinusoidal forces have different amplitudes, frequencies and sometimes phases. The reaction wheel is tested at different speeds of operation, then the results are analyzed to extract these amplitudes and frequencies. This method is very straight forward as it depends on the data obtained from the test and getting the values of the amplitudes and frequencies is not difficult. However, there are some drawbacks of this method. One of the main drawbacks is that this method depends on the tests, which means that the micro-vibrations model will be obtained after the manufacturing of the reaction wheel. This might not be the best way money wise as if the reaction wheel

* Corresponding author.
E-mail addresses: hmalkomy@yorku.ca (H. Alkomy), jjshan@yorku.ca (J. Shan).

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has micro-vibrations over the required level, it will be useless, and a new reaction wheel should be used. In some cases, one can make simple modifications to satisfy the required micro-vibration level. Because of being an experimental model, another disadvantage appears when the tested reaction wheel induces micro-vibrations higher than the desired limits. In such cases, it is not easy to suggest a specific modification to satisfy the required micro-vibration level. For example, it is very difficult to suggest reducing the bearing waviness amplitude, or any other disturbance source, by a certain value to satisfy the required micro-vibration level. As a brief overview on some of the research work that used this method, one can start with the main contribution to this method [1,2]. In these research works, a four degrees of freedom (DOFs) disturbance model was derived empirically. The radial forces and moments were studied over the range of speeds (0 – 2500) rpm and the range of frequencies (0–300) Hz. A systematic way to obtain the amplitudes and frequencies of the disturbance forces from the experimental results was presented in the paper. In [3], the coupling between the momentum wheel assembly and the mounting structure was considered. The paper followed the empirical method to model the momentum wheel micro-vibrations and tests were run to validate their proposed model. Dynamically, the momentum wheel and the reaction wheel have the same structure and their micro-vibrations have the same behavior.

The combined models can be derived by dividing the reaction wheel model into two parts: analytical part and empirical part. The concept behind this is that part of the reaction wheel disturbance model can be easily obtained analytically, whereas the other part is more difficult to be obtained analytically. For example, the static and dynamic imbalances can be represented analytically with small simplifications. In addition, considering bearings with linear stiffness facilitate the modeling of the bearing disturbances analytically. Disturbances coming from other sources inside the reaction wheel, such as the motor, or coming from model assumptions, such as the linear stiffness bearings, can be added to the model empirically. Such a method, partially, overcomes the main disadvantage of the empirical models as not the whole reaction wheel is modeled empirically. In fact, this method is practically acceptable as it is easy to be implemented and gives a general overview of the micro-vibrations behavior prior to the manufacturing process. Also, it does not require advanced mathematical models or big computational times. However, a detailed analytical model and accurate behavior expectations cannot be obtained from this method. Thus, many researchers focused on this method and a lot of research works can be found. For instance, some researchers considered a five DOF model with linear stiffness bearings and wheel imbalances [4,5]. The authors also considered the variable stiffness due to the position of the balls of the ball bearing during rotation. Other researchers studied a different structure of wheels, the cantilever type, and considered soft-suspension and wheel imbalances. In order to model the soft-suspension system, they had to model the bearing by representing the linear and rotational motions using linear and torsional springs [6–8]. Another research work focused on estimating the parameters of the combined model from the data collected form the experiments [9]. In [10], the authors investigated the coupling between a cantilever RW and the structure considering the gyroscopic effects in its accelerance. Not very different from the previous paper's main idea, authors in [11] studied the coupling between the wheel and the satellite considering a cantilever RW. They considered both cases, stiff mounting and flexible mounting on the satellite structure.

Regarding the analytical model, the main concern is that a highly nonlinear model is required. This adds significant complexity to the modeling, increases the computational time, and make the control design complicated. On the other hand, there are many benefits that can be gained from this model. On the top of these benefits, the micro-vibrations behavior can be expected before the manufacturing of the reaction wheel itself. Thus, design recommendations can be provided to design and manufacturing engineers to keep the micro-vibration level within the required limits. Also, some solutions can be suggested to decrease the micro-vibration level of manufactured reaction wheels if it does not satisfy the requirements. Due to the complexity of this method, there are not many related research work in literature. Wang et al. [12] provided a nonlinear model for a moment wheel assembly that considers a nonlinear stiffness and wheel imbalances. Although the model is for moment wheels, it can be used for reaction wheels as well. In the literature, research works related to the rotor-bearing systems can be found [13–18]. The same concepts in the aforementioned research works can be applied to the reaction wheel.

There are some papers related to reaction wheels micro-vibrations, but have a wider scope than modeling. The micro-vibrations modeling is part of these publications, but not the main objective of them. Looking into these papers, one can recognize the importance of developing an accurate dynamics model as it has a significant effect on other steps in spacecraft design. For example, some researchers conducted a system level analysis of the induced micro-vibrations and studied the effect on the pointing accuracy of the satellite [19]. The authors considered the static and dynamic imbalances only as disturbance sources in their dynamics model. Another paper dealt with the micro-vibration mitigation [20]. In terms of modeling, they used the combined model. They considered the wheel imbalances analytically, and other disturbance sources empirically. In [21], the micro-vibrations of the reaction wheel were passively isolated by mounting the reaction wheel on a low frequency flexible mounting. In [22], the authors designed an isolation system for the micro-vibrations of the momentum wheel. They validated their design by experimental tests. Another paper studied the fully coupled effect between the reaction wheel and the spacecraft [23]. They considered multiple reaction wheels and studied the effect of the reaction wheels on the spacecraft and vice versa. In their study, they considered the static and dynamic imbalances and used an analytical model. In [24], the authors developed a method to characterize a family of RWs and define a disturbance input matrix that encompasses the effects of that family. This matrix takes into account the different geometric sources of micro-vibrations of this family. The reaction wheels themselves were used as actuators to suppress the vibration of a flexible spacecraft as presented in [25]. A main source of the micro-vibrations in the reaction wheels in the bearing, so the literature related to the dynamics modeling and vibration of the contact bearings is important and some of this literature can be found in [26–28]. From a dynamical point of view, the reaction wheel is similar to rotor-bearing systems, so the publications related to the modeling of rotor-bearing systems are very useful and helpful. They provided the authors of this paper with good understanding of the problem and some of these publications can be found in [29–34].

It is obvious that deriving an analytical dynamics model of the reaction wheel that considers as many disturbance sources as possible is vital and has a significant effect on both design and manufacturing stages. Also, it is clear that this process involves highly nonlinear terms and requires a good understanding of ball bearing modeling.

In this research work, an analytical dynamics model of the reaction wheel is proposed. The model considers the static and dynamic imbalances of the wheel. Moreover, the nonlinear stiffness of the bearing is considered and modeled using the Hertz contact theory. This means that the variable stiffness has been considered and the effect of the position of each ball during rotation has been taken into account. Moreover, some bearing imperfections have been considered such as the waviness in inner and outer races, and waviness of the balls themselves. The effect of the lubrication has been added to the model by adding its damping to the equations of motion. In order to validate the model, the results have been compared with experimental results. In Section 2, the dynamic model is derived analytically including the aforementioned disturbance sources. The simulation and experimental results are presented in Section 3 and Section 4, respectively. The results are discussed and analyzed in Section 5. Section 6 concludes this paper. The main contribution of this research work is the accurate nonlinear analytical model, expressed in physical parameters, that can describe the reaction wheel's micro-vibration behavior. This model considers, analytically, the following disturbance sources: static/dynamic imbalances, bearing nonlinearity, variable stiffness and waviness in all bearing components. It also considers the damping effect of the lubricant. Another application-related contribution is that the model can be used to predict the micro-vibration behavior of the reaction wheel, check if it is within the acceptable limits, and/or provide suggestions to reduce the micro-vibration level in the design stage, *i.e.*,before the manufacturing of the reaction wheel itself. This is efficient money wise and time wise. The latter contribution is due to the accuracy of the model and the fact that it is expressed in terms of physical parameters.

2. Dynamics modeling

2.1. Static and dynamic imbalances of the wheel

In most cases, the wheel material can be considered to be homogeneously distributed over the volume of the wheel. However, in case of micro-vibrations, this assumption does not work as our concern is about micro-vibrations not vibrations. In order to identify the wheel imbalances, it is better to study the material distribution in two different cases: the plane of rotation and over the wheel thickness (planes parallel to the plane of rotation). The non-homogeneous distribution in the rotational plane causes the static imbalance; whereas the non-homogeneous distribution over the wheel thickness causes the dynamic imbalance. The static imbalance effect appears as a radial force (similar to the centrifugal force), whereas the dynamic imbalance effect appears as a moment about the radial axis. Thus, static and dynamic imbalances can be modelled as point masses at specific radii as shown in Fig. 1. In terms of the quantity, the static imbalance force can be calculated from Eq. (1), whereas the dynamic imbalance moment can be quantified using Eq. (2):

$$F_s = m_s r_s \Omega^2 \tag{1}$$

$$M_d = 2 m_d r_d d \Omega^2 \tag{2}$$

where, m_s is the static imbalance, r_s is the radius of the static imbalance, m_d is the dynamic imbalance, r_d is the radius of the dynamic imbalance and d is half of the thickness of the wheel.

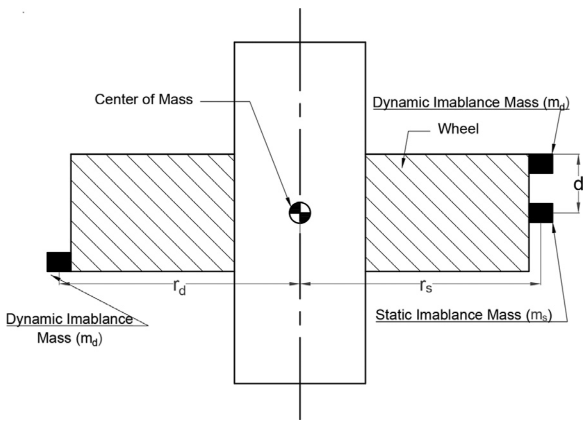


Fig. 1. Static and dynamic imbalances modeling.

2.2. Bearing stiffness and hertz contact theory

2.2.1. Hertz contact theory and contact forces

Hertz contact theory [16] is used to accurately model the ball bearing, which is the source of many disturbances. With this theory, the contact mechanics problem between two elastic bodies is solved. A general overview and the essential concepts are presented here and more details can be found in [16,18]. In general, Hertz contact theory follows the Eq. (3):

$$F = K\delta^n \tag{3}$$

where, F is the elastic force between the bodies, K is the stiffness, δ is the deflection and n is the power that depends on the type of contact. In case of point contact (such as ball bearing), $n = 1.5$.

2.2.2. Stiffness calculations

The general idea is to calculate the stiffness and the deflection of each ball inside the ball bearing, then calculate the Hertzian force resulting from this ball. After that, the balls' forces can be added up to calculate the bearing disturbance forces and moments. Consequently, all the calculations in Section 2.2.2 through Section 2.6 are for one ball unless otherwise mentioned. This way, the equations would be easier to understand, and the concept would be explained more clearly.

Prior to calculating the stiffness, some terms must be determined first. The main term to determine is the curvature difference for the Hertzian contact, $F(\rho)$, which can be calculated using Eq. (4):

$$F(\rho) = \frac{(\rho_{I1} - \rho_{I2}) + (\rho_{II1} - \rho_{II2})}{\sum \rho} \tag{4}$$

where, ρ is the curvature of one body in a specific plane, the subscripts I and II refer to the first and second body, respectively, the subscripts 1 and 2 refer to the two principle planes of the bodies and $\Sigma\rho$ is the curvature sum of the four curvatures of the two bodies. For more details, references [16,18] can be reviewed. Given the bearing material properties, the stiffness of the bearing can be calculated as in Eq. (5)

$$K = \frac{4\sqrt{2}}{3} \frac{1}{\sqrt{\sum \rho}} \frac{E_I E_{II}}{E_I + E_{II} - E_I \nu_{II}^2 - E_{II}^2} \left(\frac{1}{\delta^*}\right)^{1.5} \tag{5}$$

where, E is the Young's modulus of elasticity, ν is the Poisson's ratio and δ^* is nondimensional deflection.

The nondimensional deflection, δ^*, can be related to the curvature difference, $F(\rho)$, and the graphs for this relation can be found in [16,18]. For simulation purposes, polynomial curve fitting techniques are used to approximate these graphs into polynomial equations, which are more suitable to simulations. In order to ensure high level of accuracy in the fitting process, the graph has been divided into three portions (with three different polynomials) depending on how sharp the slope of the curve is. For the range of $0 < F(\rho) \leq 0.9$, the nondimensional deflection, δ^*, can be calculated from the polynomial in Eq. (6):

$$\delta^* = -4.473F(\rho)^6 + 10.221F(\rho)^5 - 9.138F(\rho)^4 + 3.728F(\rho)^3 - 0.912F(\rho)^2 + 0.042F(\rho) + 1 \tag{6}$$

For the range of $0.9 < F(\rho) \leq 0.995$, the nondimensional deflection, δ^*, can be calculated from the polynomial in Eq. (7):

$$\delta^* = -1.784 * 10^5 F(\rho)^5 + 8.361 * 10^5 F(\rho)^4 - 1.567 * 10^6 F(\rho)^3 + 1.469 * 10^6 F(\rho)^2 - 6.883 * 10^6 F(\rho) + 1.29 * 10^5 \tag{7}$$

For the range of $0.995 < F(\rho) < 1$, the nondimensional deflection, δ^*, can be calculated from the polynomial in Eq. (8):

$$\begin{aligned}\delta^* = &-2.839 * 10^{11} F(\rho)^5 + 1.414 * 10^{12} F(\rho)^4 - 2.817 * 10^{12} F(\rho)^3 \\ &+ 2.806 * 10^{12} F(\rho)^2 - 1.398 * 10^{12} F(\rho) + 2.784 * 10^{11}\end{aligned} \tag{8}$$

2.3. Bearing waviness model

Bearing waviness is an unavoidable imperfection as it comes from the machining process or even during bearing operation. Thus, it is important to consider its effect on the induced micro-vibrations. It is acceptable to describe the waviness as a sinusoidal wave [17,30]. Waviness may exist on all bearing surfaces (balls and inner/outer races) and in all directions (radial and axial) [12,14,17,30]. Fig. 2 is a schematic drawing that illustrates the waviness in bearing components in axial and radial directions. Fig. 3 shows a general wave representation on any bearing component, while Eq. (9) indicates its mathematical representation.

$$w = A \sin\left[2\pi \frac{(j-1)}{N_b} - N_w(\omega_{race} - \omega_c)t\right] \tag{9}$$

where, w is the amplitude of the waviness at any time t, A is the maximum amplitude of the waviness, N_b is the total number of balls of the bearing, j is the number of the ball in contact with the bearing race, N_w the waviness order, ω_{race}

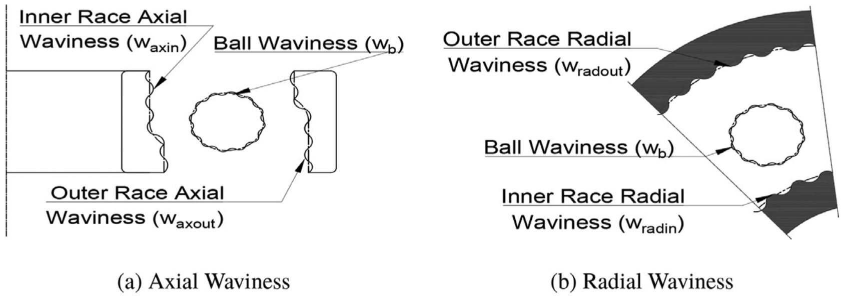


Fig. 2. Schematic drawing for the bearing waviness.

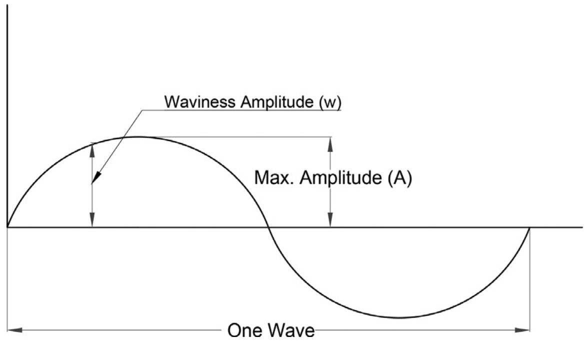


Fig. 3. General wave representation.

is the angular velocity of the bearing races, and ω_c is the angular velocity of the cage. In reality, the balls are free to spin about any axis, but a simple and acceptable case is to consider that the ball rotates about an axis perpendicular to the plane containing the centers of the inner and outer race contacts. In this case, Eq. (10) gives the instantaneous amplitude of the waviness

$$w = A\sin(N_w\omega_b t) \tag{10}$$

where, ω_b is the angular spinning velocity of the ball.

2.3.1. Bearing inner and outer races waviness

The presented model differentiates between the effect of radial and axial waviness. This is the first step to study the effect of each kind of waviness individually. The subscripts (*radin*) and (*axin*) are used to represent the radial and axial term of the inner race, respectively, while the subscripts (*radout*) and (*axout*) are used to represent the radial and axial term of the outer race, respectively. Eq. (11) shows the radial waviness of the inner race, whereas Eq. (12) illustrates the axial waviness of the inner race

$$w_{radin} = A_{radin}\sin\left[2\pi\frac{(j-1)}{N_b} - N_{wradin}(\omega_{in}-\omega_c)t\right] \tag{11}$$

$$w_{axin} = A_{axin}\sin\left[2\pi\frac{(j-1)}{N_b} - N_{waxin}(\omega_{in}-\omega_c)t\right] \tag{12}$$

Eq. (13) shows the radial waviness of the outer race, whereas Eq. (14) illustrates the axial waviness of the outer race

$$w_{radout} = A_{radout}\sin\left[2\pi\frac{(j-1)}{N_b} - N_{wradout}(\omega_{out}-\omega_c)t\right] \tag{13}$$

$$w_{axout} = A_{axout}\sin\left[2\pi\frac{(j-1)}{N_b} - N_{waxout}(\omega_{out}-\omega_c)t\right] \tag{14}$$

2.3.2. Bearing ball race waviness

For the ball waviness, a phase difference of angle π should be considered between inner and outer races. Consequently, the ball waviness can be represented as in Eq. (15). The subscript (b) is used to represent terms related to the ball

$$w_b = 2A_b\sin(N_{wb}\omega_b t) \tag{15}$$

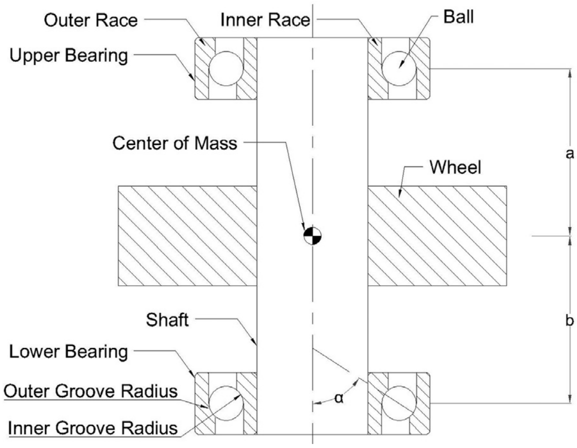


Fig. 4. Schematic drawing of the reaction wheel assembly.

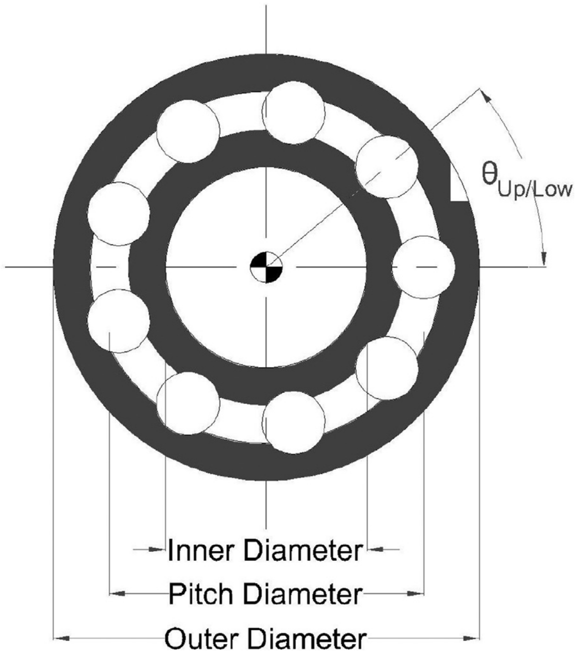


Fig. 5. Bearing main dimensions.

2.4. Deflection calculations

In order to calculate the deflection correctly, many factors should be considered. The linear and rotational rigid body motion of the wheel itself should be considered. The elastic deflection due to the waviness is vital in the calculation process. Due to the deflection, the initial contact angle changes with time. In fact, it even changes from one ball to another. Figs. 4 and 5 show the reaction wheel assembly and bearing dimensions, respectively, to help specify the deflection. The deflection can be analyzed into two components: radial deflection and axial deflection. The axial component can be found from Eqs. (16) and (17), while the radial component can be calculated from Eq. (18) and Eq. (19). The total deflection can be found from Eqs. (20) and (21). The subscripts (*Up*) and (*Low*) refer to the upper and lower bearings, respectively.

$$\delta_{axUp} = H\sin(\alpha) + Z_o - z + w_{axin} + w_{axout} + \frac{d_m}{2}[\theta_y\cos(\theta_{Up}) + \theta_x\sin(\theta_{Up})] \tag{16}$$

$$\delta_{axLow} = H\sin(\alpha) + Z_o - z + w_{axin} + w_{axout} + \frac{d_m}{2}[\theta_y\cos(\theta_{Low}) + \theta_x\sin(\theta_{Low})] \tag{17}$$

$$\delta_{radUp} = H\cos(\alpha) + \left[x + y + a\sin(\theta_y)\cos\left(\theta_{Up}\right) + a\sin(\theta_x)\right]\sin\left(\theta_{Up}\right) + w_{radin} + w_{radout} \tag{18}$$

$$\delta_{radLow} = H\cos(\alpha) + [x + y - b\sin(\theta_y)\cos(\theta_{Low}) + b\sin(\theta_x)]\sin(\theta_{Low}) + w_{radin} + w_{radout} \tag{19}$$

$$\delta_{Up} = \sqrt{\delta_{axUp}^2 + \delta_{radUp}^2} - H + w_b \tag{20}$$

$$\delta_{Low} = \sqrt{\delta_{axLow}^2 + \delta_{radLow}^2} - H + w_b \tag{21}$$

where, the azimuth angle of the jth ball in the upper bearing $\theta_{Up} = 2\pi\frac{(j-1)}{N_b} + \omega_c t + \gamma_{Up}$ and the azimuth angle of the jth ball in the lower bearing $\theta_{Low} = 2\pi\frac{(j-1)}{N_b} + \omega_c t + \gamma_{Low}$. Without loss of generality, the phase angle, γ, can be assumed to be zero. x, y, z, θ_x and θ_y are the position and orientation of the center of mass of the reaction wheel. Z_o is the axial deflection due to the bearing preload. The distance between the centers of the inner and outer grooves is H, while a and b are the axial distance between the center of mass of the reaction wheel and the center of the upper and lower bearings respectively. One can identify these parameters from Fig. 4. The pitch diameter of the bearing is d_m and can be found in Fig. 5.

Consequently, the instant contact angle of each ball can be calculated from Eqs. (22) and (23).

$$\alpha_{Up} = \tan^{-1}\frac{\delta_{axUp}}{\delta_{radUp}} \tag{22}$$

$$\alpha_{Low} = \tan^{-1}\frac{\delta_{axLow}}{\delta_{radLow}} \tag{23}$$

2.5. Damping effect

The main source of damping inside the ball bearing is the lubricant, which has a damping force proportional to the velocity [15,16,18]. In case of ball bearings used in the reaction and momentum wheels, a value of 300 N.s/m for the damping coefficient, c is acceptable. It is possible to have different damping coefficients in the radial and axial directions, c_r and c_a. So it is considered in the derivation of the equations of motion. Practically, the difference is very small so that it is reasonable to assume that the damping coefficients are the same in both directions.

2.6. Equations of motion

In order to derive the equations of motion, some terms and parameters need to be defined first. The subscripts (*Up*) and (*Low*) refer to the upper and lower bearing, respectively. The force of each bearing has three components and can be calculated from Eqs. (24)–(29).

$$F_{xUp} = F_{Up}\cos\left(\alpha_{Up}\right)\cos(\theta_{Up}) \tag{24}$$

$$F_{xLow} = F_{Low}\cos(\alpha_{Low})\cos(\theta_{Low}) \tag{25}$$

$$F_{yUp} = F_{Up}\cos\left(\alpha_{Up}\right)\sin(\theta_{Up}) \tag{26}$$

$$F_{yLow} = F_{Low}\cos(\alpha_{Low})\sin(\theta_{Low}) \tag{27}$$

$$F_{zUp} = F_{Up}\sin\left(\alpha_{Up}\right) \tag{28}$$

$$F_{zLow} = F_{Up}\sin\left(\alpha_{Up}\right) \tag{29}$$

The moments of each bearing about the two radial axes can be determined from Eqs. (30)–(33).

$$M_{xUp} = a\ F_{Up}\cos\left(\alpha_{Up}\right)\sin\left(\theta_{Up}\right) + \frac{d_m}{2}F_{Up}\sin\left(\alpha_{Up}\right)\sin\left(\theta_{Up}\right) \tag{30}$$

$$M_{xLow} = b\ F_{Low}\cos(\alpha_{Low})\sin(\theta_{Low}) + \frac{d_m}{2}F_{Low}\sin(\alpha_{Low})\sin(\theta_{Low}) \tag{31}$$

$$M_{yUp} = a\ F_{Up}\cos\left(\alpha_{Up}\right)\cos\left(\theta_{Up}\right) + \frac{d_m}{2}F_{Up}\sin\left(\alpha_{Up}\right)\cos\left(\theta_{Up}\right) \tag{32}$$

$$M_{yLow} = b\ F_{Low} \cos(\alpha_{Low}) \cos(\theta_{Low}) + \frac{d_m}{2} F_{Low} \sin(\alpha_{Low}) \cos(\theta_{Low}) \tag{33}$$

The presented model considers the reaction wheel as a five DOFs system. The nonlinear equations of motion can be written in a short matrix form as follows:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} = \mathbf{F}_{unbalance} - \mathbf{F}_{bearings} \tag{34}$$

where $\mathbf{q}$, $\mathbf{M}$, $\mathbf{C}$, $\mathbf{F}_{unbalance}$, and $\mathbf{F}_{bearings}$ are the generalized coordinates, the inertia matrix, the damping matrix, the vector of the wheel unbalance forces and moments, and the bearings forces and moments vector, respectively. These vectors and matrices can be defined as follows:

$$\mathbf{q} = \begin{bmatrix} x \\ y \\ z \\ \theta_x \\ \theta_y \end{bmatrix}, \mathbf{M} = \begin{bmatrix} M & 0 & 0 & 0 & 0 \\ 0 & M & 0 & 0 & 0 \\ 0 & 0 & M & 0 & 0 \\ 0 & 0 & 0 & I_r & 0 \\ 0 & 0 & 0 & 0 & I_r \end{bmatrix}, \mathbf{C} = \begin{bmatrix} c_r & 0 & 0 & 0 & 0 \\ 0 & c_r & 0 & 0 & 0 \\ 0 & 0 & c_a & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z\Omega \\ 0 & 0 & 0 & -I_z\Omega & 0 \end{bmatrix},$$

$$\mathbf{F}_{unbalance} = \begin{bmatrix} -F_s \sin(\Omega t) \\ F_s \cos(\Omega t) \\ 0 \\ M_d \cos(\Omega t) \\ M_d \sin(\Omega t) \end{bmatrix}, \mathbf{F}_{bearings} = \begin{bmatrix} F_{xUp} + F_{xLow} \\ F_{yUp} + F_{yLow} \\ F_{zLow} - F_{zUp} \\ M_{xLow} - M_{xUp} \\ M_{yUp} - M_{yLow} \end{bmatrix},$$

where, I_r and I_z are the mass moments of inertia in the radial and axial direction, respectively.

3. Simulation results

The equations of motion have been solved numerically using Runge-Kutta 4th-5th order method. The flowchart in Fig. 6 shows the solution procedure.

Table 1 shows the parameters used in simulation. For the sake of meaningful analysis, the results have been analyzed in the engine order domain. From a vibration point of view, the engine order analysis helps quantify the vibration of rotating machinery whose rotational speed varies. Basically, an engine order represents the multiplication of the rotational frequency. This multiplication factor can be any positive rational number. Mathematically, the engine order is the ratio between the disturbance frequency and the rotational frequency. Consequently, the engine order domain is kind of a normalized frequency domain that facilitates identifying the disturbance sources as any disturbance source appears as a series of spikes that has the same engine order at all rotational speeds.

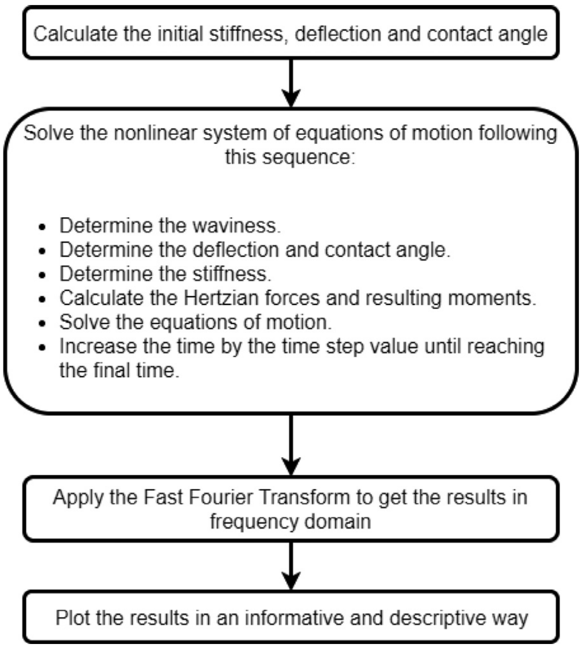


Fig. 6. Solution procedure.

Table 1
Simulation parameters.

Parameter	Value	Unit
Wheel mass	0.55	kg
Radial moment of inertia	$233.215e-6$	$kg \cdot m^2$
Polar moment of inertia	$185.968e-6$	$kg \cdot m^2$
Static wheel imbalance	$0.01e-6$	kg
Dynamic wheel imbalance	$0.012e-6$	kg
Bearing free contact angle	15	deg
Bearing inner diameter	12.6	mm
Bearing outer diameter	17.5	mm
Bearing ball diameter	3.969	mm
Bearing inner groove radius	4.08	mm
Bearing outer groove radius	4.61	mm
Number of balls in each bearing	9	
Elastic modulus of elasticity	$2.06e11$	N/m^2
Poisson's ratio	0.33	
Inner waviness maximum amplitude (radial and axial)	$1e-6$	m
Outer waviness maximum amplitude (radial and axial)	$5e-6$	m
Ball waviness maximum amplitude	$1e-6$	m
Speed range	$0-6000$	rpm

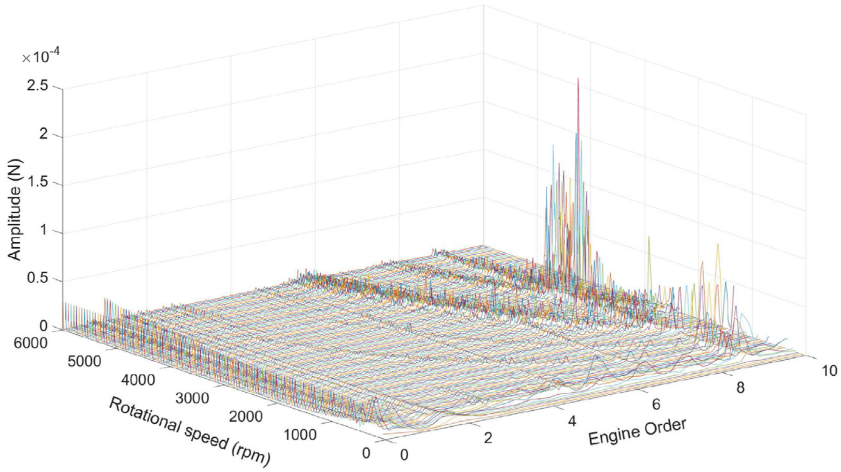


Fig. 7. Simulation results of disturbance forces in X-direction.

The forces in three dimensions and the moments about the radial axis are shown as waterfall plots in Figs. 7–11. In each plot, the force/moment is plotted versus the engine order at different rotational speeds. In simulations, the speed changes from 0 to 6000 rpm at 50 rpm step. The range and the step are selected to be the same as those used in experiments.

In Figs. 7–11 the disturbance sources appear as line of spikes parallel to the rotational speed axis, i.e., Y-axis. The imbalance disturbance source occurs at engine order equal to 1. The outer race waviness disturbance source can be found at different engine orders based on the waviness order. In this simulation, waviness order 1 and 3 are considered and their main disturbance sources occur, approximately, at the engine orders 3 and 8.9, respectively. The inner race waviness of order 1 is considered in this simulation and its disturbance occurs at an engine order, approximately, equal to 5. A ball waviness with a waviness order equal to 1 results in a main disturbance source at engine order, approximately, equal to 7. Other disturbances can be found at other engine orders and these can be originated from the interference between the different disturbance sources. In addition, some disturbance sources have main engine order and some other side engine orders such as the imbalance, which has a main engine order at 1 and side engine orders at 2, 3, ... etc. Random spikes that do not for a line parallel to Y-axis do not represent a disturbance source. In simulation results, they can come from numerical errors and approximations.

4. Experimental studies

4.1. Experimental setup

Fig. 12 shows a plan view of the tested reaction wheel mounted on the Kistler table used in the experimental study. The used table is Kistler 9255C, which has four piezoelectric force sensors that can measure the force in three orthogo-

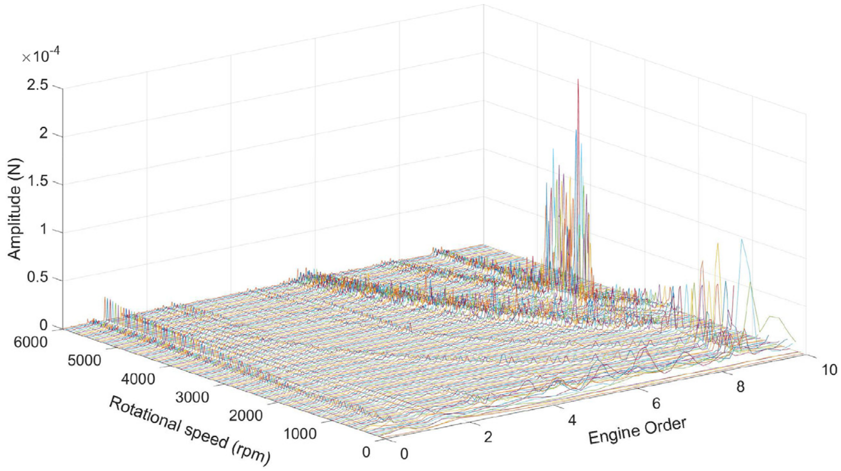


Fig. 8. Simulation results of disturbance forces in *Y*-direction.

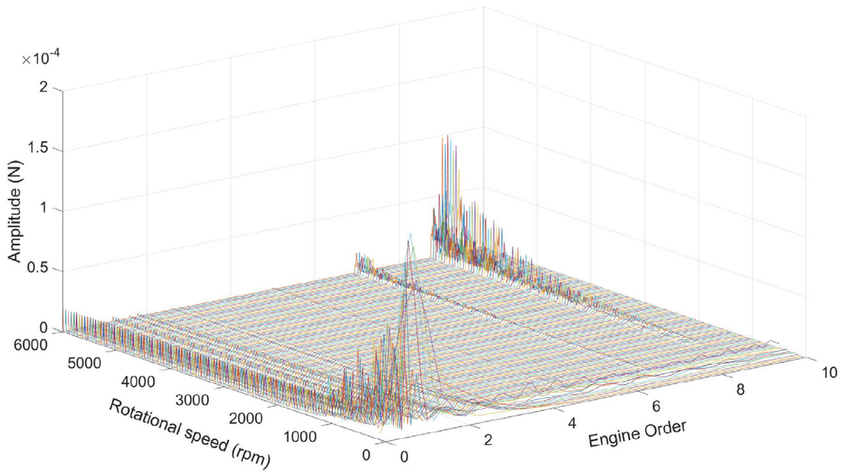


Fig. 9. Simulation results of disturbance forces in *Z*-direction.

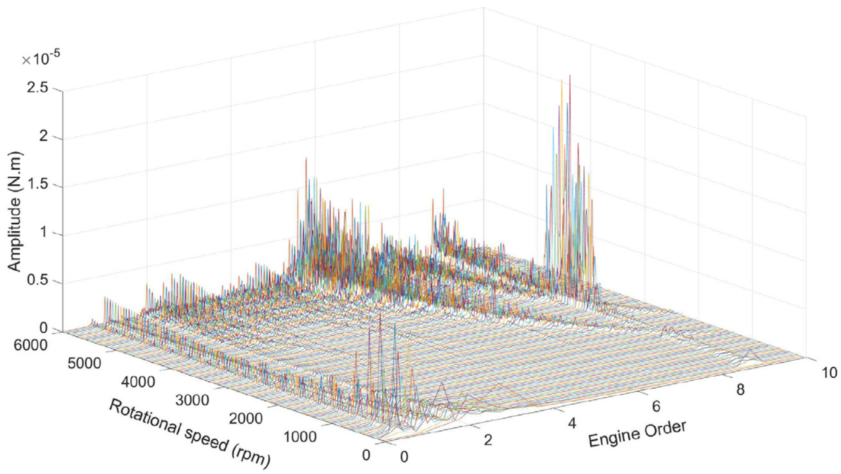


Fig. 10. Simulation results of disturbance moments about *X*-axis.

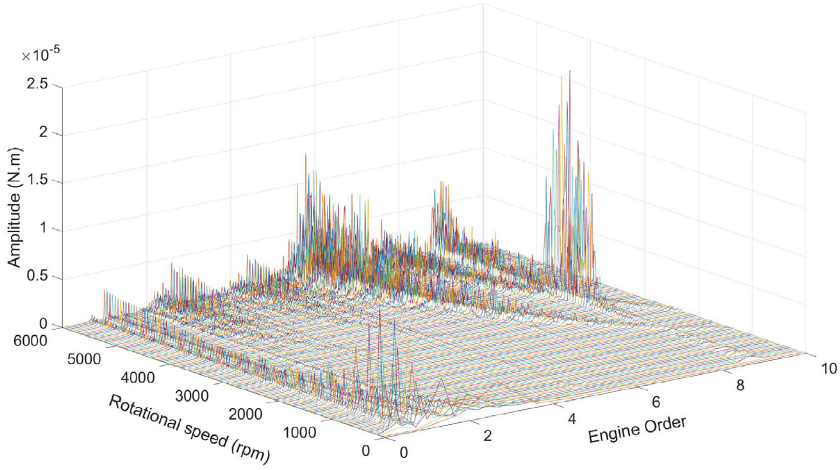


Fig. 11. Simulation results of disturbance moments about Y-axis.

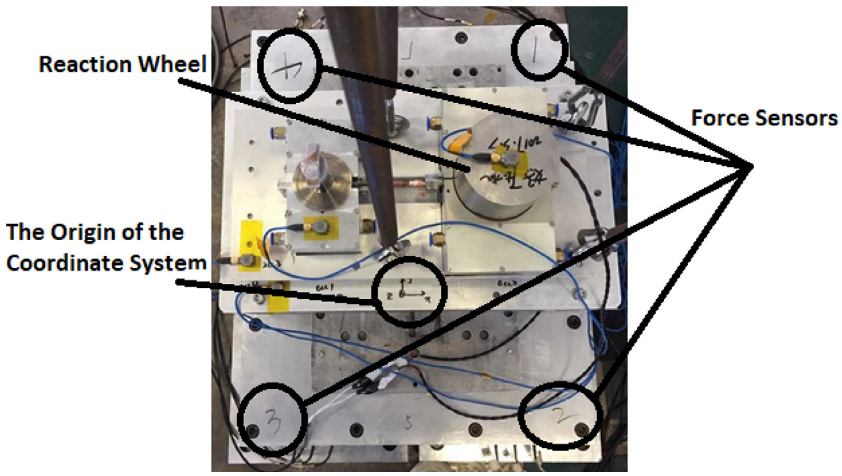


Fig. 12. Kistler table for experimental study.

nal dimensions. Kistler 9255C can measure radial forces, i.e., in x and/or y directions, within the range of $\pm$ 30 kN with a sensitivity of about −7.9 pC/N, whereas it can measure the axial force, i.e., in z direction, within the range of −10 to 60 kN with a sensitivity of about −3.9 pC/N. The dimensions of the table and locations of the forces sensors are shown in Fig. 13. The reaction wheel along with Kistler 9255C, were mounted on an air bearing table to reduce the transmitted disturbances from the ground. The air bearing table is shown in Fig. 14 and it has an area of 1.8 × 1.6 m^2. The efficiency of the vibration isolation is more than 80% at a frequency of 5 Hz. The reaction wheel was controlled to spin at different rotational speeds. It spun at speeds that range from 0 to 6000 rpm with a step of 50 rpm. The force sensors' measurements at each speed were logged via the 24-bit data acquisition (LMS SC316-UTP). Using this extracted data, the total disturbance forces in three directions and the total disturbance moments about the radial axis can be determined using Eq. (35).

$$\begin{aligned} F_x &= F_{x1} + F_{x2} + F_{x3} + F_{x4} \\ F_y &= F_{y1} + F_{y2} + F_{y3} + F_{y4} \\ F_z &= F_{z1} + F_{z2} + F_{z3} + F_{z4} \\ M_x &= l_y(F_{z1} - F_{z2} - F_{z3} + F_{z4}) \\ M_y &= l_x(-F_{z1} - F_{z2} + F_{z3} + F_{z4}) \end{aligned} \tag{35}$$

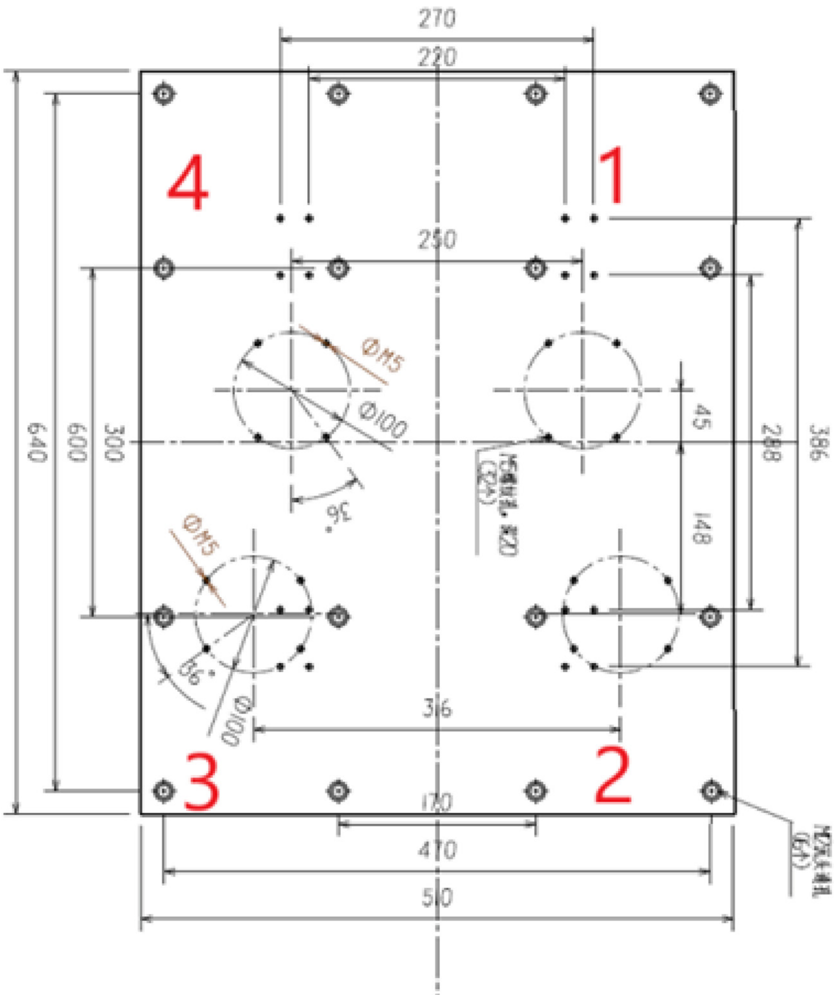


Fig. 13. The dimensions of Kistler table with the locations of the four force sensors.

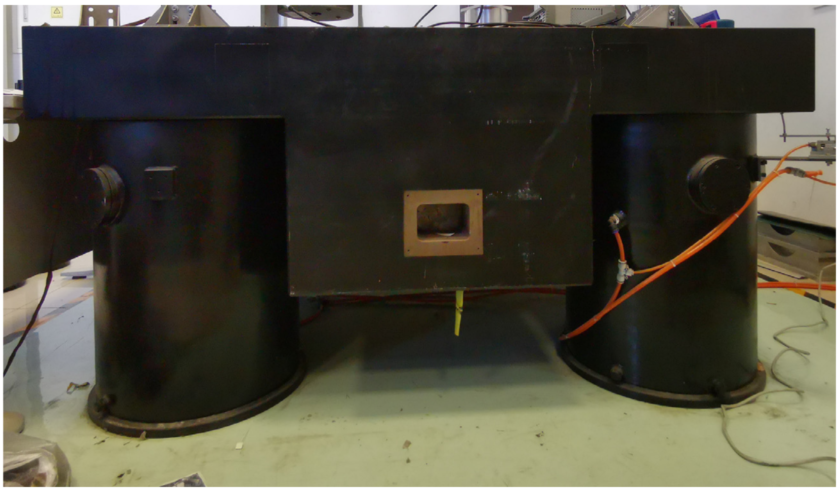
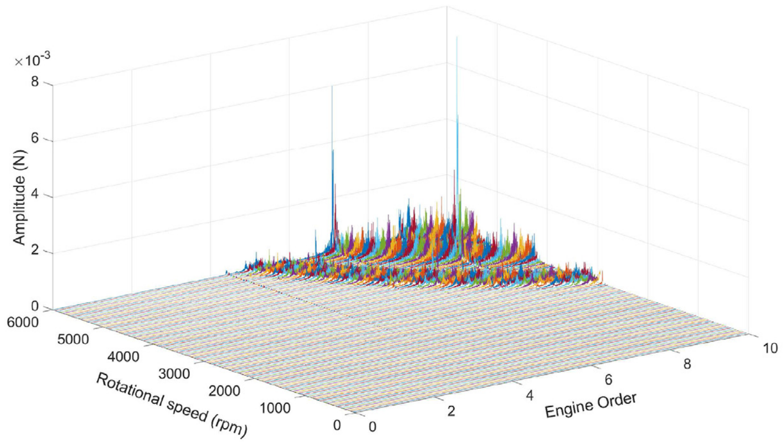
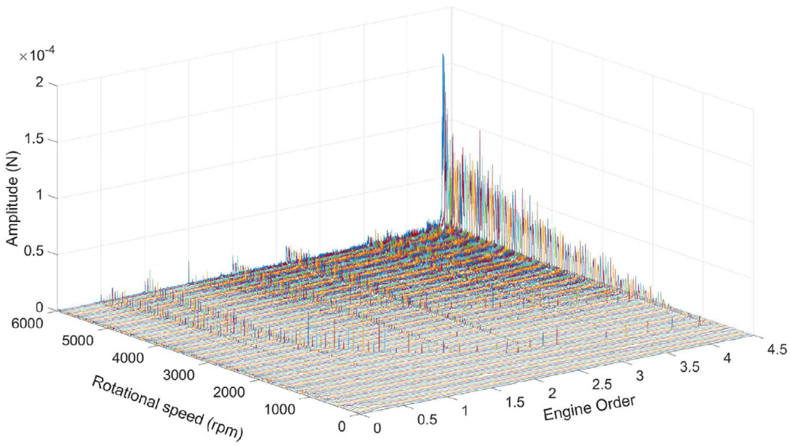


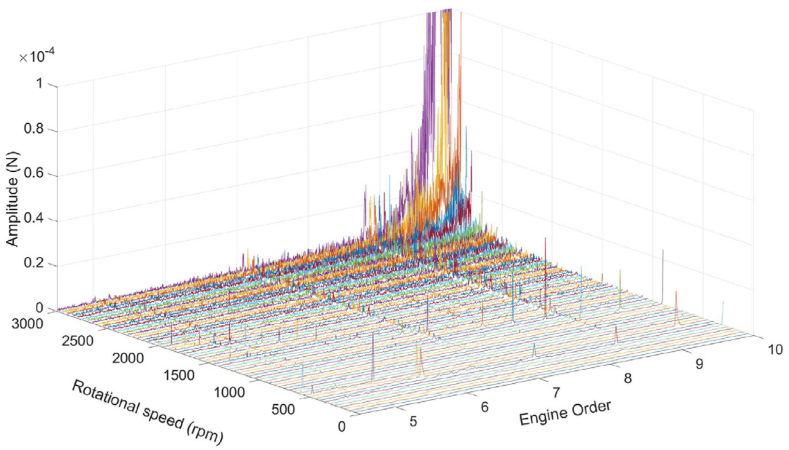
Fig. 14. The air bearing table.



(a) Disturbance Forces in X-direction

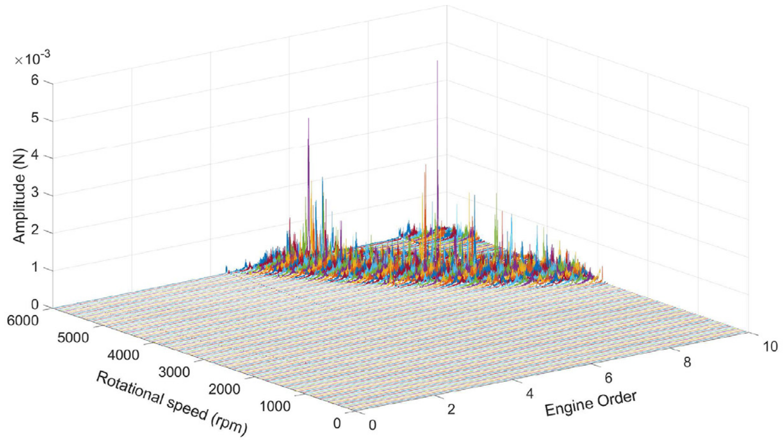


(b) Disturbance Forces in X-direction Engine Orders (0-4.5)

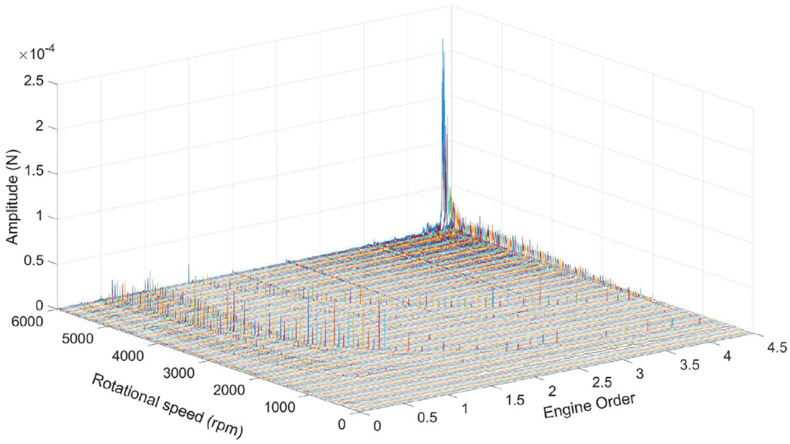


(c) Disturbance Forces in X-direction Engine Orders (4.5-10)

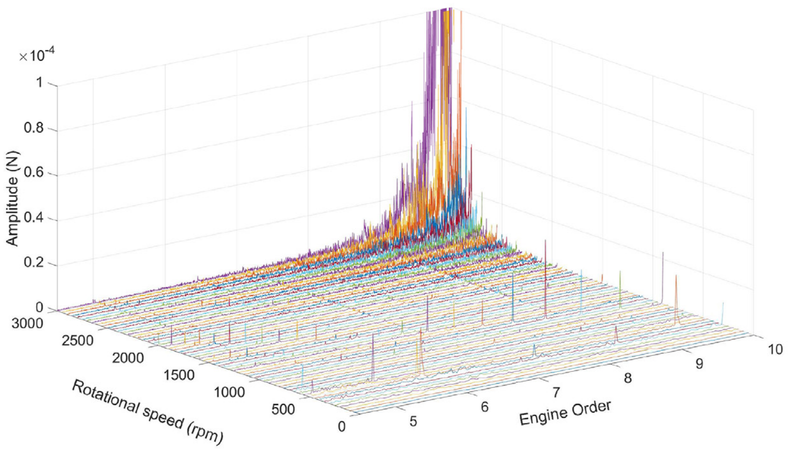
Fig. 15. Experimental results of disturbance forces in X-direction.



(a) Disturbance Forces in Y-direction

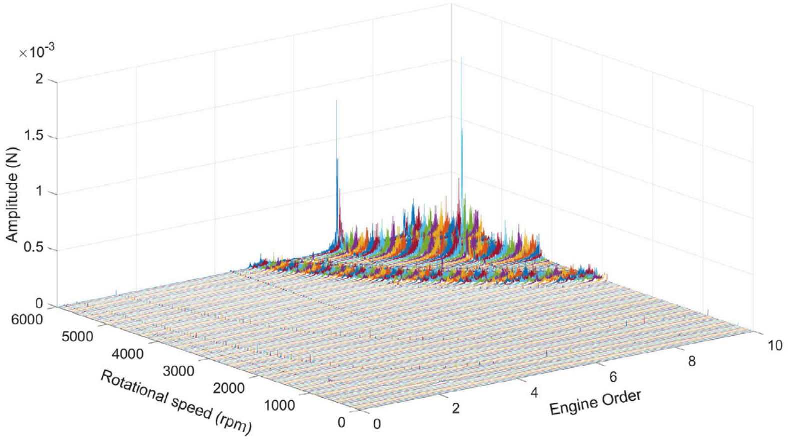


(b) Disturbance Forces in Y-direction Engine Orders (0-4.5)

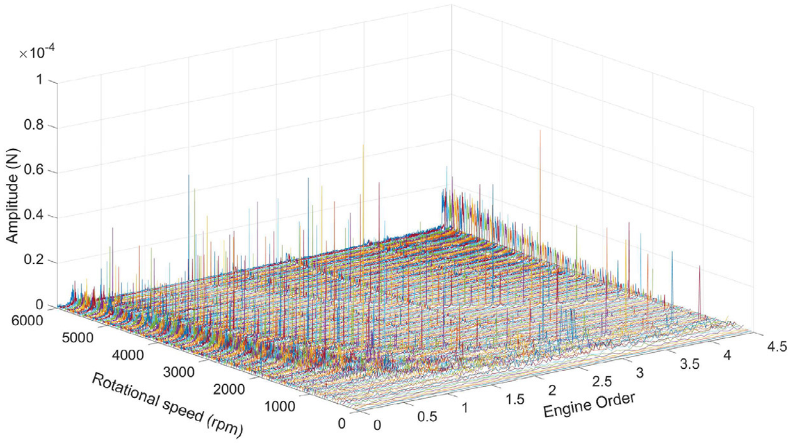


(c) Disturbance Forces in Y-direction Engine Orders (4.5-10)

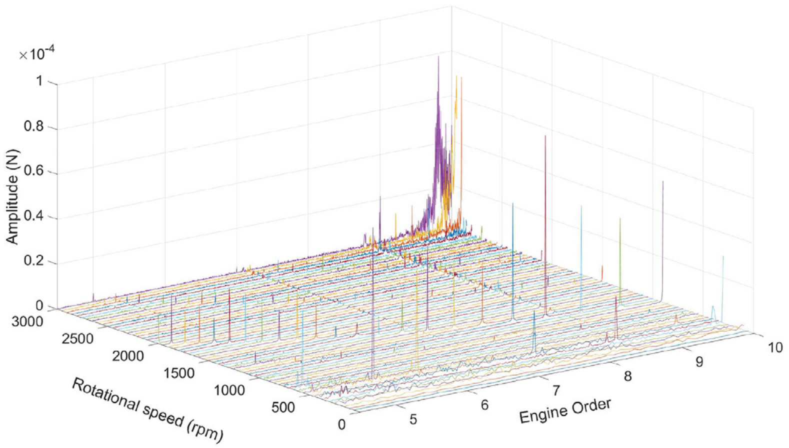
Fig. 16. Experimental results of disturbance forces in *Y*-direction.



(a) Disturbance Forces in Z-direction

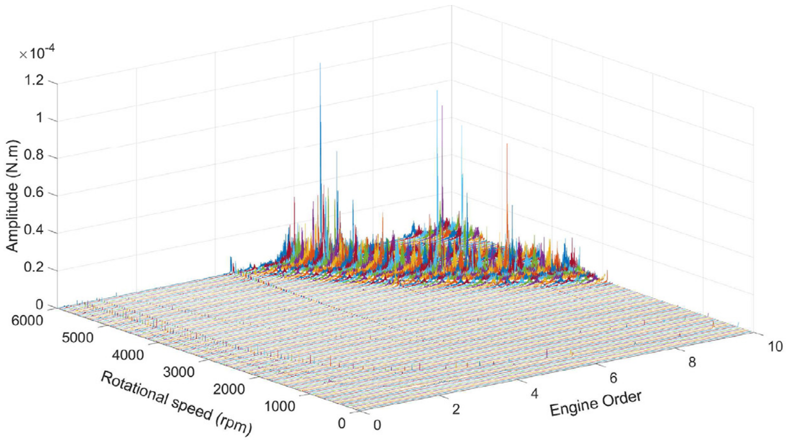


(b) Disturbance Forces in Z-direction Engine Orders (0-4.5)

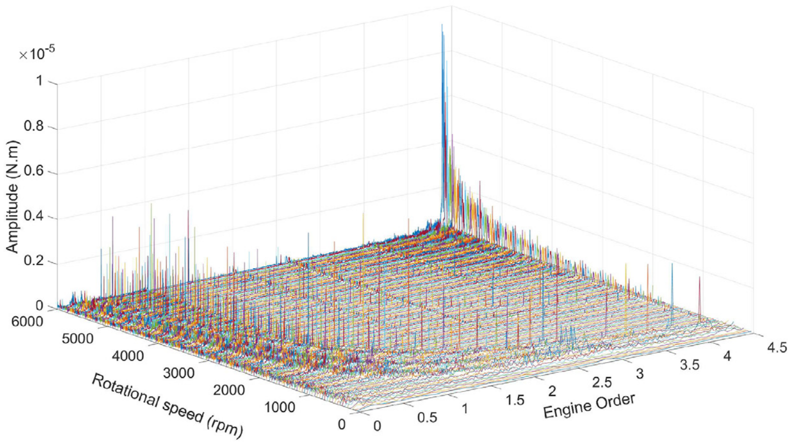


(c) Disturbance Forces in Z-direction Engine Orders (4.5-10)

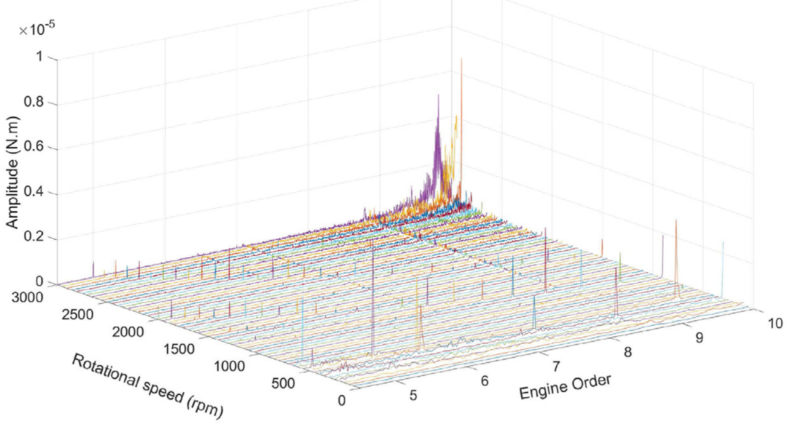
Fig. 17. Experimental results of disturbance forces in Z-direction.



(a) Disturbance Moments about X-axis

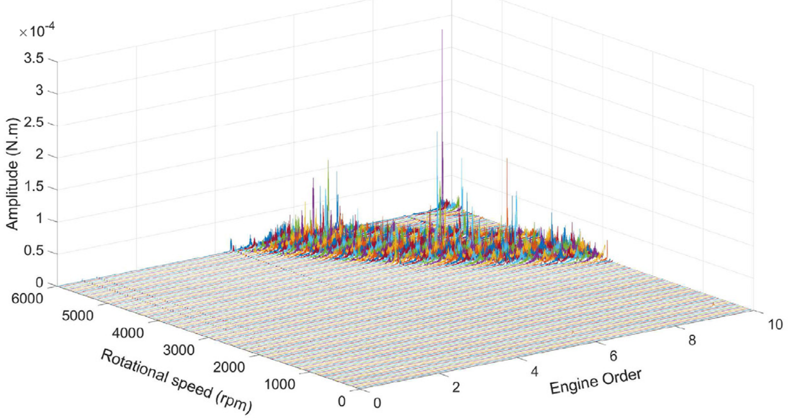


(b) Disturbance Moments about X-axis Engine Orders (0-4.5)

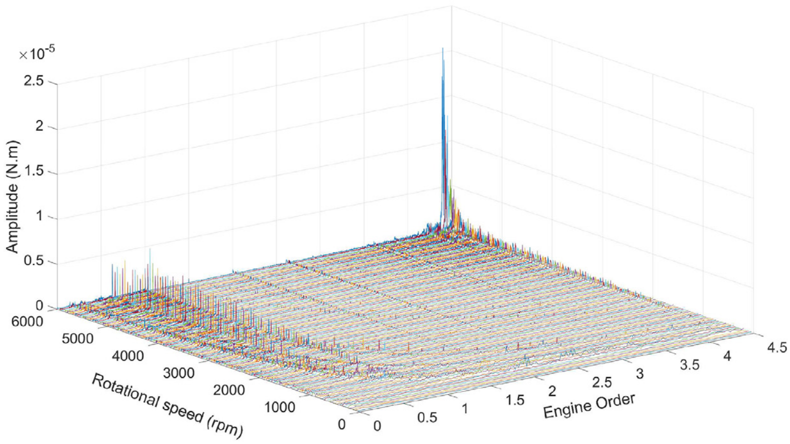


(c) Disturbance Moments about X-axis Engine Orders (4.5-10)

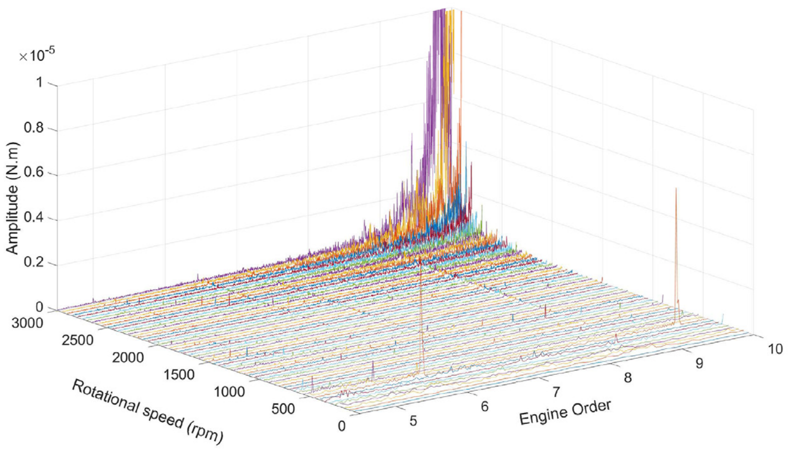
Fig. 18. Experimental results of disturbance moments about *X*-axis.



(a) Disturbance Moments about Y-axis

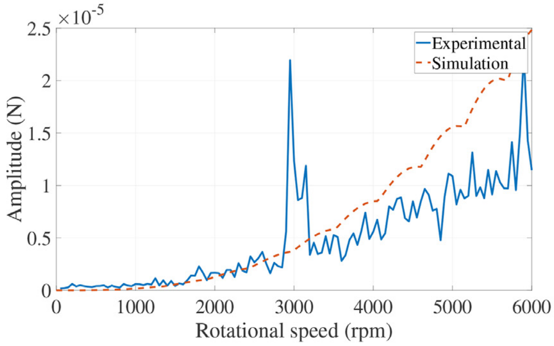


(b) Disturbance Moments about Y-axis Engine Orders (0-4.5)

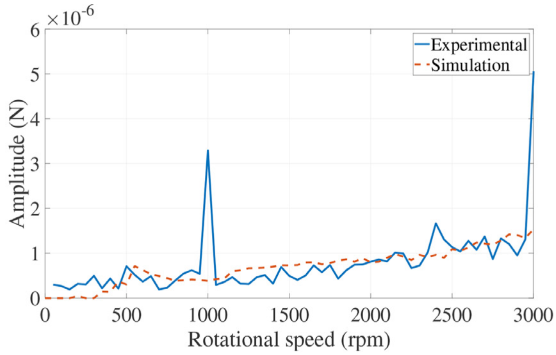


(c) Disturbance Moments about Y-axis Engine Orders (4.5-10)

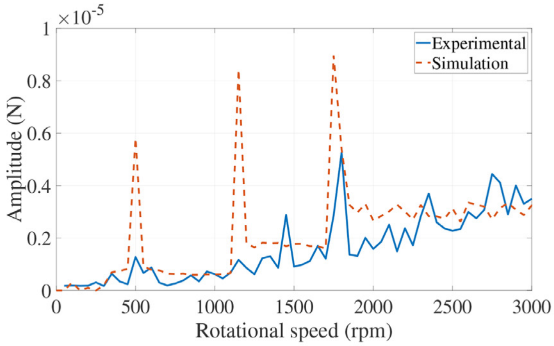
Fig. 19. Experimental results of disturbance moments about *Y*-axis.



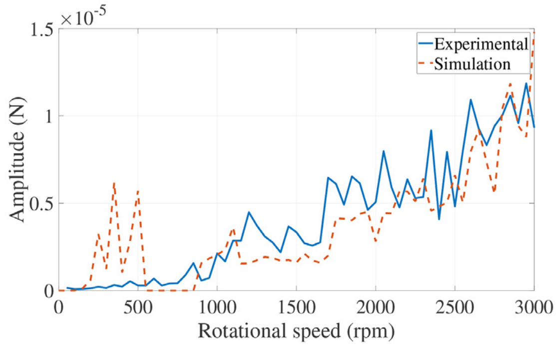
(a) Imbalance Disturbance in X-direction (N) vs Rotational speed (rpm)



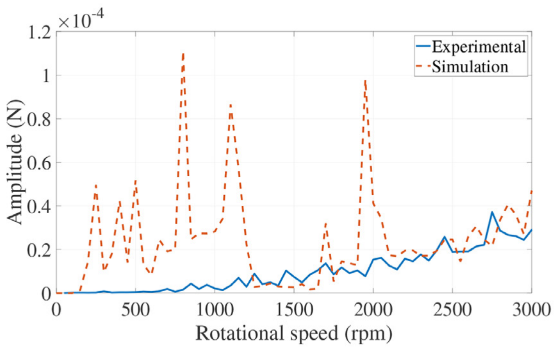
(b) Outer Race Disturbance [Order 1] in X-direction (N) vs Rotational speed (rpm)



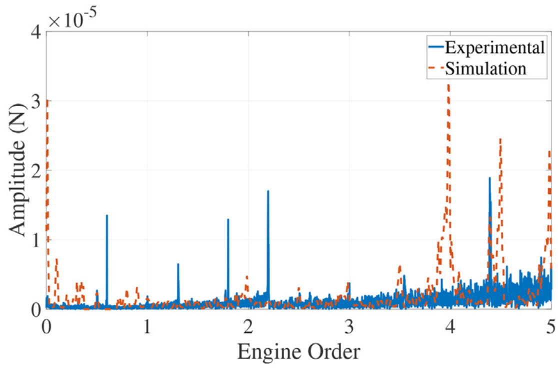
(c) Inner Race Disturbance in X-direction (N) vs Rotational speed (rpm)



(d) Ball Disturbance in X-direction (N) vs Rotational speed (rpm)



(e) Outer Race Disturbance [Order 3] in X-direction (N) vs Rotational speed (rpm)

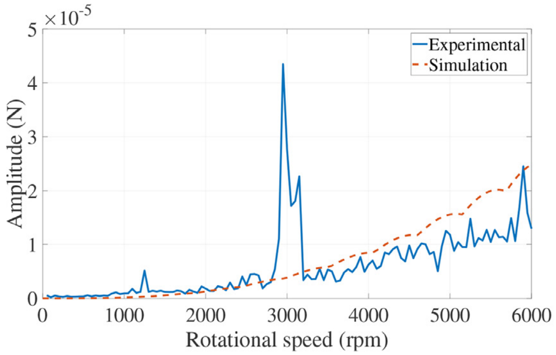


(f) All Disturbance Forces in X-direction (N) vs Engine Order at 2500 rpm

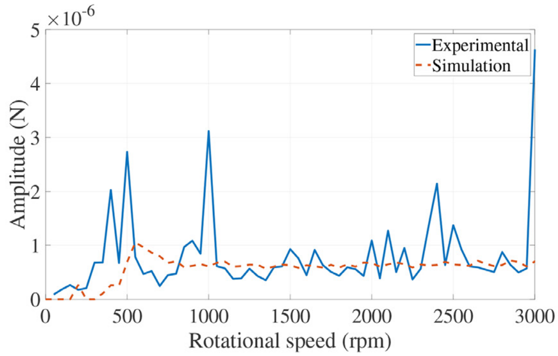
Fig. 20. Radial disturbance forces in *X*-direction.

4.2. Experimental results

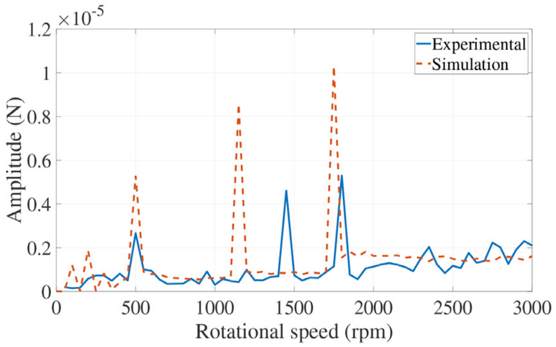
The experimental results have been plotted the same way as the simulation results, i.e. in the engine order domain. The same speed range, 0 to 6000 rpm, with a step of 50 rpm has been used. There are two limitations that have affected the experimental results. The first limitation is the natural frequency of the equipment itself, which is 500 Hz. This means that the micro-vibrations with frequencies larger than 500 Hz will be affected by the resonance and the measurements cannot be accurate. The second limitation is the sampling frequency of the force sensor, which is 2048 Hz and, practically, accurate measurements are guaranteed at 25% of this sampling frequency. This means that the accurate measurements can be obtained for micro-vibrations with frequencies lower than 512 Hz, approximately. These two limitations make the results



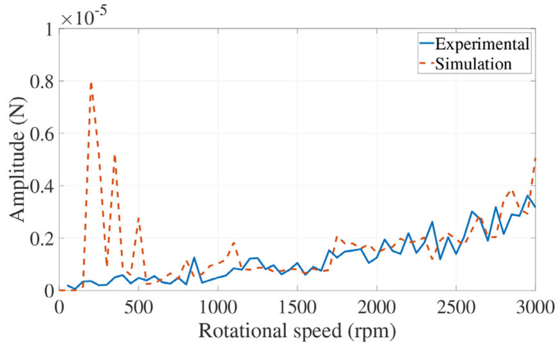
(a) Imbalance Disturbance in Y-direction (N) vs Rotational speed (rpm)



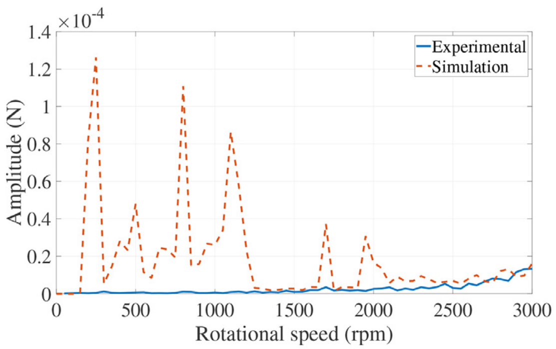
(b) Outer Race Disturbance [Order 1] in Y-direction (N) vs Rotational speed (rpm)



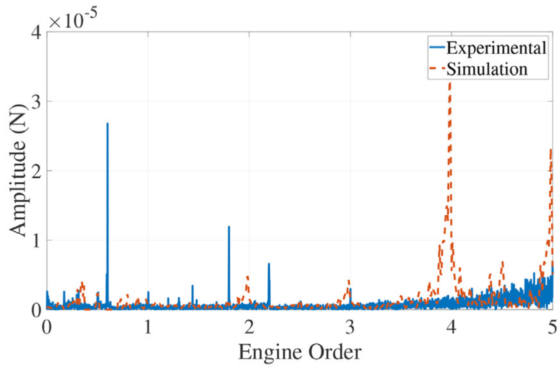
(c) Inner Race Disturbance in Y-direction (N) vs Rotational speed (rpm)



(d) Ball Disturbance in Y-direction (N) vs Rotational speed (rpm)



(e) Outer Race Disturbance [Order 3] in Y-direction (N) vs Rotational speed (rpm)

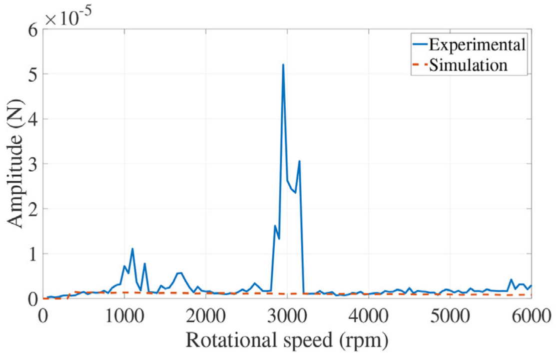


(f) All Disturbance Forces in Y-direction (N) vs Engine Order at 2500 rpm

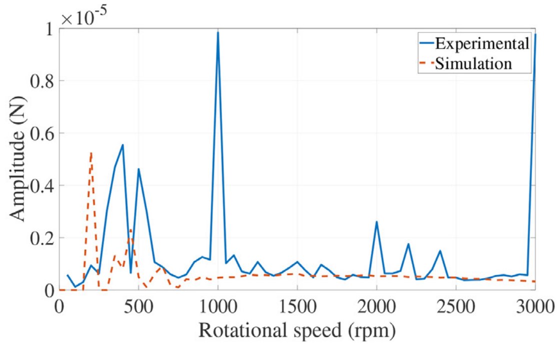
Fig. 21. Radial disturbance forces in *Y*-direction.

obtained for higher engine orders at higher rotational speeds inaccurate and cannot be validated. For this reason, the way of presenting the experimental results is a little bit different from that used to show the simulation results.

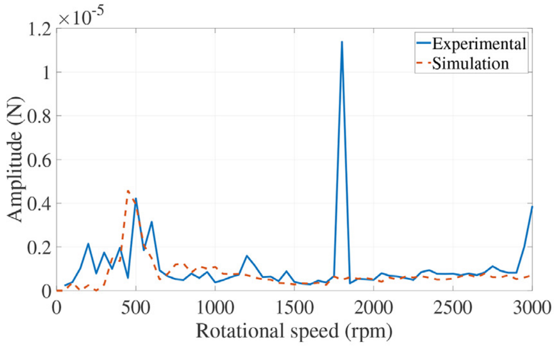
The forces in three dimensions and the moments about the radial axis are shown as waterfall plots in Figs. 15–19. In each waterfall graph, the force/moment is plotted versus the engine order at different rotational speeds. The difference between presenting the simulation and experimental results is that in case of experimental results each force/moment has three subfigures. Figure (a) is to show the entire range of engine orders from 1 to 10 and one can see the effect of the resonance. Figure (b) is a detailed plot focusing on engine orders from 1 to 4.5. This is the range of engine orders that is not affected by the resonance. After engine order $= 4.5$, the effect of the resonance appears at higher rotational speeds. Consequently, Figure (c) illustrates the details of the higher engine order at rotational speeds up to 3000 rpm only. This is to avoid the effect of the resonance at higher speeds.



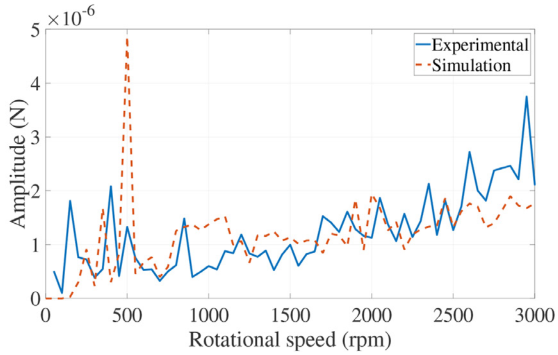
(a) Imbalance Disturbance in Z-direction (N) vs Rotational speed (rpm)



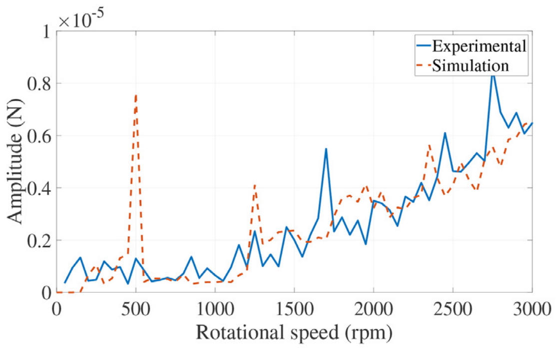
(b) Outer Race Disturbance [Order 1] in Z-direction (N) vs Rotational speed (rpm)



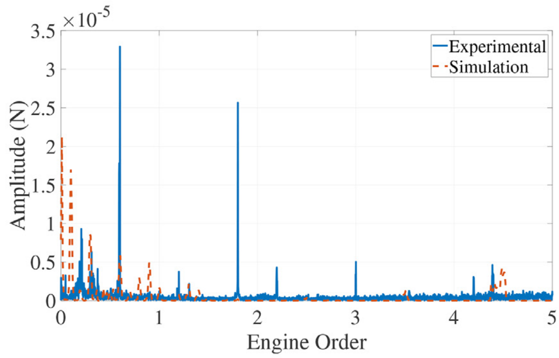
(c) Inner Race Disturbance in Z-direction (N) vs Rotational speed (rpm)



(d) Ball Disturbance in Z-direction (N) vs Rotational speed (rpm)



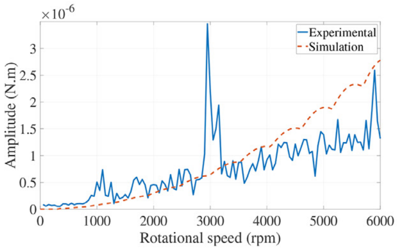
(e) Outer Race Disturbance [Order 3] in Z-direction (N) vs Rotational speed (rpm)



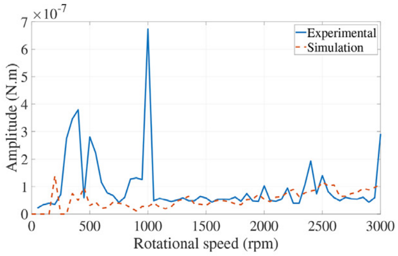
(f) All Disturbance Forces in Z-direction (N) vs Engine Order at 2500 rpm

Fig. 22. Axial disturbance forces in *Z*-direction.

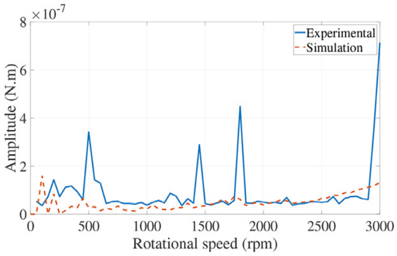
Figs. 15–19 represents the disturbance sources that appears in the experiment. Similar to the simulation results in Section 3, the main disturbance sources have the same engine orders. This means, in brief, that the imbalance disturbance occurs at engine order equal to 1, whereas the outer race waviness disturbances occur, approximately, at the engine orders 3 and 8.9 for waviness orders 1 and 3, respectively. The inner race waviness disturbance occurs at an engine order, approximately, equal to 5, while the ball waviness disturbance occurs at engine order, approximately, equal to 7. Similar to the simulation results in Figs. 7–11, other disturbances can be found in the experimental results in Figs. 15–19. In addition to the reasons stated in the simulation results such as the interference between the different disturbance sources, the unmodeled disturbance sources appear and interfere with other sources in the experimental results. That is why one can notice more disturbances in the experimental results than in the simulation results. Again, random spikes that do not for a line parallel to *Y*-axis do not represent a disturbance source. In experimental results, the reason behind these random spikes can be some sensor or environment noise.



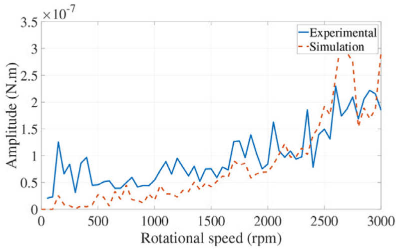
(a) Imbalance Disturbance Moment about X-direction (N.m) vs Rotational speed (rpm)



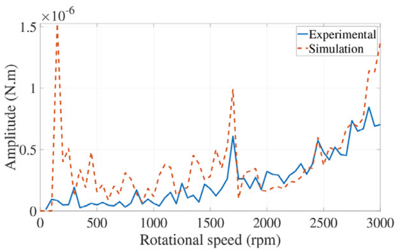
(b) Outer Race Disturbance Moment [Order 1] about X-direction (N.m) vs Rotational speed (rpm)



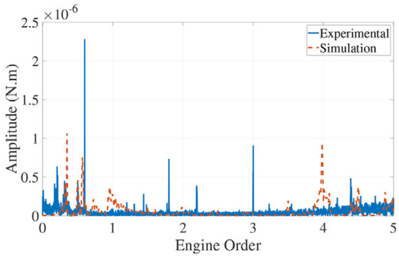
(c) Inner Race Disturbance Moment about X-direction (N.m) vs Rotational speed (rpm)



(d) Ball Disturbance Moment about X-direction (N.m) vs Rotational speed (rpm)



(e) Outer Race Disturbance Moment [Order 3] about X-direction (N.m) vs Rotational speed (rpm)



(f) All Disturbance Moments about X-direction (N.m) vs Engine Order at 2500 rpm

Fig. 23. Disturbance moments about *X*-axis.

5. Discussions

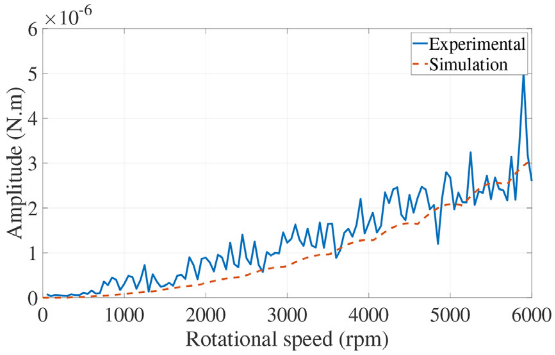
It is worth mentioning that the experimental results include all kinds of the actual disturbance sources, whereas the simulation results are limited to the considered disturbance sources. Therefore some of the disturbance sources in the experimental results cannot be found in the simulation results.

Comparisons between the simulation and experimental results have been done for each degree of freedom including the imbalance disturbance, the inner race waviness disturbance, the ball waviness disturbance and the outer race waviness disturbance at two waviness orders. In addition, a full disturbance comparison is conducted at 2500 rpm, which is the middle of the test speed.

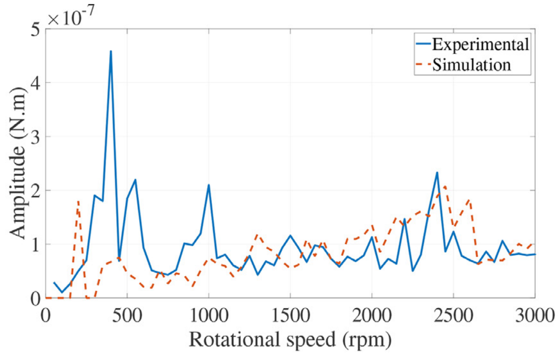
Generally, one can see some spikes that are inconsistent in simulation and experimental results. For the experimental results, the main reason is that the unavoidable interference between the micro-vibration signals may magnify or reduce the resultant amplitude. Moreover, the noise can cause some of these unexpected spikes. The subfigures (b) and (c) in all figures in Section 4.2 illustrate this point and of course these spikes appear in all figures in Section 5. For the simulation results, the sudden spikes come mostly from the interference between the micro-vibrations and from error accumulation in the solution procedure. The most important thing is that the simulation results agree well with the experimental results at the moderate and high speed range, where the micro-vibrations effect is significant.

5.1. Radial force in X-direction

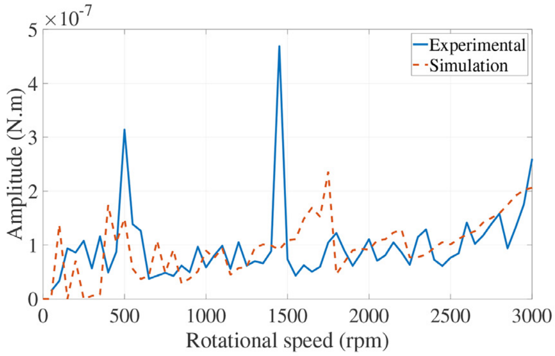
Fig. 20 shows the comparison between simulation and experimental results in *X*-direction. Figs. 20a–e show the radial disturbance force in *X*-direction due to the static imbalance and bearing disturbances at different speeds. Fig. 20f illustrates the entire frequency response of the radial force in *X*-direction at the speed of 2500 rpm. The proposed model can represent the disturbance forces in the *X*-direction. A good agreement with the experimental result, especially at moderate and high speeds, can be proved by comparing the results. Fig. 20e shows large spikes in simulation at low speeds (lower than 1250



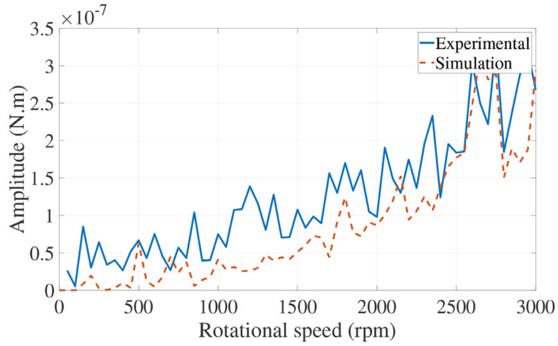
(a) Imbalance Disturbance Moment about Y-direction (N.m) vs Rotational speed (rpm)



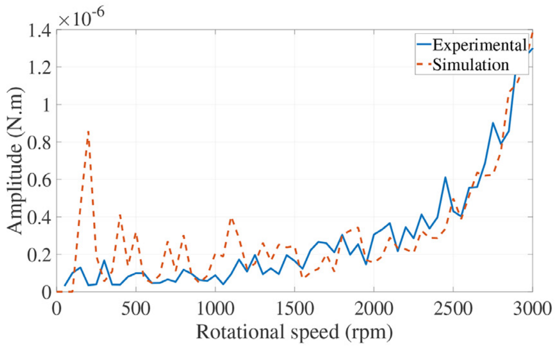
(b) Outer Race Disturbance Moment [Order 1] about Y-direction (N.m) vs Rotational speed (rpm)



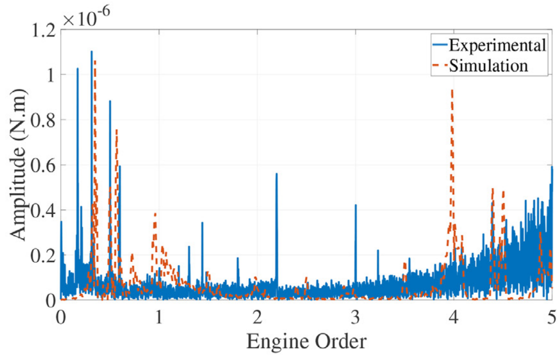
(c) Inner Race Disturbance Moment about Y-direction (N.m) vs Rotational speed (rpm)



(d) Ball Disturbance Moment about Y-direction (N.m) vs Rotational speed (rpm)



(e) Outer Race Disturbance Moment [Order 3] about Y-direction (N.m) vs Rotational speed (rpm)



(f) All Disturbance Moments about Y-direction (N.m) vs Engine Order at 2500 rpm

Fig. 24. Disturbance moments about *Y*-axis.

rpm). As mentioned in the introduction of Section 5, this might come from the interference between the frequency of this disturbance source (outer race waviness of order 3) and other unconsidered disturbances.

5.2. Radial force in Y*-direction*

Similarly, Fig. 21 illustrates the comparison between simulation and experimental results in *Y*-direction. Figs. 21a–e indicate the radial disturbance force in *Y*-direction due to the static imbalance and bearing disturbances at different speeds. Fig. 21f shows the entire frequency response of the radial force in *Y*-direction at the speed of 2500 rpm. Again, Fig. 21e shows large spikes in simulation at low speeds (lower than 1250 rpm). For speeds higher than 1250 rpm, good agreement can be found. As explained in Section 5 and Section 5.1, the unconsidered interference between disturbance sources frequencies could be the reason for these spikes. Otherwise, good agreement can be found for the forces in the *Y*-direction for the rest of the disturbance sources.

5.3. Axial force in Z-direction

Following the same procedure, Fig. 22 is a comparison between simulation and experimenatl results. Figs. 22a–e show the axial disturbance force in the *Z*-direction due to the static imbalance and bearing disturbances at different speeds. Fig. 22f indicates the entire frequency response of the radial force in *Z*-direction at the speed of 2500 rpm.

5.4. Moment about X-axis

In case of moment about the *X*-axis, the agreement between the simulation and experimental results can be found in Fig. 23. The disturbance moments of each disturbance source is illustrated by Figs. 23a–e. Fig. 23f shows the entire frequency response of the moment about the *X*-axis at the speed of 2500 rpm.

5.5. Moment about Y-axis

Similarly, Fig. 24 shows the simulation and experimental results of the disturbance moments around the *Y*-axis. Figs. 24a–e illustrate the agreement of the proposed model by comparing the simulation results with the experimental ones for each disturbance source. Again, Fig. 24f shows the entire frequency response of the moment about the *Y*-axis at the speed of 2500 rpm.

6. Conclusions

In this paper, an analytical nonlinear five-degree-of-freedom dynamics model is proposed for the micro-vibrations of the reaction wheel (RW). Different from most of the models in literature, which depends on empirical parameters to an extent, the proposed model depends on physical parameters that can be measured before and/or after the manufacturing of the reaction wheel. These physical parameters include RW mass and dimensions, wheel imbalance measurements, bearing dimensions and material properties, and bearing waviness measurements. The proposed model considers, analytically, the wheel static and dynamic imbalances and bearing disturbance sources. The bearing disturbance sources include the waviness in the inner and outer races in addition to the balls themselves. Experimental tests have been conducted to validate the proposed dynamics model. The results showed that the proposed dynamics model can model the micro-vibration radial and axial forces, i.e., in *X*, *Y* and *Z* directions, and rocking moments about *X* and *Y* axes. The model results have good agreement with the experimental results in moderate and high rotational speeds, i.e., higher than 700 rpm up to 3000 rpm. This agreement shows the effectiveness of the proposed model in representing the disturbance forces and moments from these different sources.

The accuracy of the proposed model in addition to the fact that it depends on physical parameters that can be measured before and/or after the manufacturing of the reaction wheel introduce some main advantages of the proposed model. Firstly, the proposed model can help reduce the cost of the manufacturing of the reaction wheels and space systems. This is because the proposed model can be used in the pre-manufacturing steps to confirm that the micro-vibrations are within the required limits. This can be achieved by inputting the physical parameters of the reaction wheel into the proposed model and run the simulation on the computer and find if the simulated micro-vibrations are within the required limits. Thanks to the accuracy of the proposed model, one can trust the results of the simulation. Secondly, if the simulated results showed that the micro-vibrations are over the required limits, modifications to reduce the level of the micro-vibrations can be suggested using the proposed model before the manufacturing of the reaction wheel. These modifications include, for instance, changing the waviness orders and/or amplitudes of the bearing races and balls. This change will be made on the computers before creating the actual reaction wheel. One can change as many parameters as desired in simulation until the designers are sure that the micro-vibration level is within the required limits. Thirdly, in case of already having the reaction wheel manufactured and it is required to reduce its micro-vibration level, one can input the physical parameters of the existing reaction wheel into the proposed model and change the parameters in simulation until the desired micro-vibration limit is met. Accordingly, some parts of the reaction wheel can be replaced by others with different parameters based on the simulation results. For example, the desired micro-vibration limit can be satisfied by changing the bearing with another one with different waviness order or with different number of balls. To sum up the main advantages of the proposed model, we can say that the accuracy and the facts that the proposed model is analytical and based on physical parameters that can be measured before or after the manufacturing of the reaction wheel allow for changing these physical parameters directly on the computers and see their reflection on the micro-vibration behavior. The reaction wheel designers can make sure that the micro-vibrations satisfy the requirements before manufacturing, suggest some modifications in case of exceeded limits of micro-vibrations. and examine the effect of these modifications just by simulation before or after the manufacturing of the reaction wheel.

Declaration of Competing Interest

The authors declare that they do not have any financial or nonfinancial conflict of interests.

CRediT authorship contribution statement

Hassan Alkomy: Conceptualization, Methodology, Writing - original draft, Visualization, Investigation, Validation. **Jinjun Shan:** Supervision, Project administration, Writing - original draft.

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